

Fleet Performance Analysis

A Data Analytics Project Proposal

For a Zimbabwean Trucking Company

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Driving

Profitability Through Data-Driven Decisions

Table of Contents

1	Executive Summary	2
2	Business Context	2
3	Project Objectives	2
3.1	Fleet Performance Analysis	3
3.2	Driver Productivity Assessment	3
3.3	Route Optimization	3
3.4	Fuel Cost Reduction	3
3.5	Maintenance Strategy	3
3.6	Financial Performance	3
3.7	Predictive Analytics	3
4	Data Requirements	3
4.1	Datasets to be Generated	3
4.2	Data Generation Method	4
5	Methodology	4
5.1	Tools and Technologies	5
6	Deliverables	5
7	Analysis Framework	5
7.1	Ten Sections of Analysis	5
8	Proposed Timeline	6
9	Skills Demonstrated	7
10	Expected Outcomes	7
11	Conclusion	8

1 Executive Summary

This proposal outlines a comprehensive data analytics project for a trucking company operating across Zimbabwe, with routes connecting Harare, Bulawayo, Mutare, Gweru, Masvingo, Beitbridge, and Chirundu.

The project aims to analyze fleet performance, driver productivity, route profitability, fuel efficiency, and maintenance costs. By leveraging descriptive analytics, statistical analysis, machine learning, and interactive dashboards, this project will deliver actionable insights to improve operational efficiency and profitability.

The deliverables include a clean dataset, SQL queries, Jupyter notebooks with analysis, machine learning models, and a Power BI dashboard that enables management to monitor key performance indicators in real time.

2 Business Context

The trucking company operates a fleet of vehicles that transport goods between major Zimbabwean cities. Management has observed rising operational costs and wants to understand how the fleet is performing to identify areas for improvement.

The company faces several challenges:

- **Fuel Costs:** Fuel represents the largest operating expense
- **Maintenance:** Unplanned repairs cause downtime and revenue loss
- **Driver Productivity:** Inconsistent performance across drivers
- **Route Efficiency:** Some routes may be unprofitable
- **Fleet Utilization:** Underperforming trucks need identification

This project will provide data-driven answers to these challenges, enabling management to make informed decisions about fleet operations.

3 Project Objectives

The primary objectives of this project are:

3.1 Fleet Performance Analysis

Evaluate each truck's performance based on revenue, trip completion, distance traveled, fuel consumption, fuel efficiency, maintenance costs, profitability, downtime, and operating cost per kilometer.

3.2 Driver Productivity Assessment

Measure driver performance by deliveries completed, revenue generated, distance driven, delays incurred, fuel efficiency, maintenance costs, average speed, profitability, and idle time.

3.3 Route Optimization

Identify which routes are most profitable, which generate losses, and where additional trucks should be allocated.

3.4 Fuel Cost Reduction

Analyze fuel consumption patterns, identify waste, and quantify potential savings from improved fuel efficiency.

3.5 Maintenance Strategy

Determine which trucks require frequent maintenance, estimate downtime costs, and recommend replacement strategies.

3.6 Financial Performance

Calculate profit margins, operating costs, and identify the company's top cost drivers.

3.7 Predictive Analytics

Build machine learning models to predict maintenance needs, fuel consumption, delivery delays, monthly revenue, and trip duration.

4 Data Requirements

4.1 Datasets to be Generated

The following datasets will be created to support the analysis:

Dataset	Key Fields	Records
Trucks	Truck ID, Model, Year, Capacity, Mileage, Purchase Date	30-50 records
Drivers	Driver ID, Name, License, Experience, Hire Date	40-60 records
Trips	Trip ID, Truck ID, Driver ID, Route, Distance, Duration, Fuel, Revenue, Cargo Weight	500-2,000 records
Maintenance	Maintenance ID, Truck ID, Date, Type, Cost, Downtime	200-500 records
Fuel Purchases	Fuel ID, Truck ID, Date, Litres, Cost, Location	300-700 records

Table 1: [Datasets for Fleet Performance Analysis](#)

4.2 Data Generation Method

Synthetic data will be generated using Python libraries (Faker, NumPy, Pandas) with realistic correlations built between variables:

- **Distance Fuel Consumption:** Linear relationship with random variation
- **Cargo Weight Fuel Efficiency:** Heavier loads reduce fuel efficiency
- **Truck Age Maintenance Costs:** Older trucks incur higher maintenance
- **Driver Experience Delays:** Experienced drivers have fewer delays

5 Methodology

The project follows a structured data analytics workflow:

1. **Data Generation:** Create synthetic datasets using Python
2. **Data Cleaning:** Handle missing values, outliers, and inconsistencies
3. **Exploratory Data Analysis (EDA):** Understand distributions and relationships
4. **SQL Analysis:** Write queries to answer the 100 business questions
5. **Statistical Analysis:** Identify significant patterns and correlations
6. **Machine Learning:** Build predictive models (Linear Regression, Random Forest, XG-Boost)
7. **Dashboard Development:** Create interactive Power BI dashboard
8. **Recommendations:** Translate findings into actionable business advice

Tool	Purpose	Why
Python	Data generation, EDA, ML	Flexible, powerful, industry standard
Jupyter Notebook	Analysis, visualization, reporting	Interactive, shareable output
SQLite	Data storage, querying	Lightweight, supports SQL
Power BI	Dashboard creation	Industry standard, executive-friendly
Scikit-learn	Machine learning models	Simple, interpretable, robust
Matplotlib/Seaborn	Statistical visualizations	High-quality plots
Pandas	Data manipulation	Essential for data wrangling
NumPy	Numerical computation	Underpins all data operations

Table 2: [Technology Stack Overview](#)

5.1 Tools and Technologies

6 Deliverables

Deliverable	Description
1. Cleaned Datasets	Raw and processed data in CSV/SQLite formats
2. SQL Queries	Comprehensive queries answering all 100 business questions
3. Jupyter Notebooks	EDA, statistical analysis, and machine learning models
4. Power BI Dashboard	Interactive dashboard with key metrics and filters
5. Machine Learning Models	Predictive models for maintenance, fuel, delays, revenue, and trip duration
6. Executive Summary	PDF report highlighting key findings and recommendations
7. GitHub Repository	Complete project with documentation and code

Table 3: [Project Deliverables](#)

7 Analysis Framework

7.1 Ten Sections of Analysis

Section	Focus
1. Fleet Overview	Overall performance metrics
2. Truck Performance	Individual truck evaluation
3. Driver Performance	Driver productivity measurement
4. Route Analysis	Best and worst performing routes
5. Fuel Analysis	Fuel consumption patterns and savings
6. Maintenance Analysis	Repair frequency, costs, and reliability
7. Financial Performance	Profitability, margins, cost drivers
8. Time-Based Analysis	Trends, seasonality, utilization
9. Predictive Analytics	ML models for forecasting
10. Recommendations	Actionable business advice

Table 4: [Analysis Framework Overview](#)

8 Proposed Timeline

Phase	Activities	Duration
Phase 1	Data generation and cleaning	2 days
Phase 2	EDA and SQL queries (Sections 1-8)	4 days
Phase 3	Machine learning models (Section 9)	3 days
Phase 4	Power BI dashboard development	2 days
Phase 5	Recommendations (Section 10) & reporting	2 days
Phase 6	Documentation & GitHub upload	1 day
Total Duration		14 days

Table 5: [Project Timeline](#)

9 Skills Demonstrated

This project will demonstrate the following competencies:

Skill Category	Specific Skills
Data Preparation	Data generation, cleaning, wrangling, handling missing values
SQL	Joins, aggregations, CTEs, window functions, subqueries
Statistical Analysis	Correlation, trend analysis, hypothesis testing
Machine Learning	Regression, classification, feature engineering, model evaluation
Data Visualization	Plotly, Matplotlib, Seaborn, Power BI dashboards
Business Intelligence	KPI definition, dashboard design, executive reporting
Communications	Translating technical findings to business recommendations
Project Management	End-to-end project delivery, version control, documentation

Table 6: Skills Demonstrated

10 Expected Outcomes

The project will produce the following insights:

1. **Identification of Best/Worst Trucks:** Clear ranking based on multiple KPIs
 2. **Driver Performance Ranking:** Data-backed evaluation for incentives or training
 3. **Route Profitability Analysis:** Which routes to expand, optimize, or discontinue
 4. **Fuel Cost Savings:** Quantified savings from 5% fuel efficiency improvement
 5. **Maintenance Cost Reduction:** Recommendations for preventive maintenance
 6. **Predictive Maintenance:** Identify trucks at risk of breakdown
 7. **Revenue Forecasting:** Monthly revenue predictions for budgeting
 8. **Operational Improvements:** Recommended changes to increase profitability
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11 Conclusion

This project provides a comprehensive analytics solution for a Zimbabwean trucking company. By combining descriptive analytics, machine learning, and interactive dashboards, it equips management with the tools needed to make data-driven decisions.

The project is structured to deliver maximum business value while demonstrating a broad range of data analytics skills. The deliverables will serve as a strong portfolio piece showcasing end-to-end analytical capabilities.

Let Data Drive Your Fleet to Greater Profitability

Appendix: Business Questions by Section

A complete list of all 100 business questions is available in the project documentation. Key highlights include:

- **Section 1:** 10 fleet overview metrics
- **Section 2:** 10 truck performance metrics
- **Section 3:** 10 driver performance metrics
- **Section 4:** 10 route analysis questions
- **Section 5:** 10 fuel analysis questions
- **Section 6:** 10 maintenance analysis questions
- **Section 7:** 10 financial performance questions
- **Section 8:** 10 time-based analysis questions
- **Section 9:** 10 predictive analytics questions
- **Section 10:** 10 management recommendations

— End of Proposal —